



SOLAR PV SYSTEMS AT PRIME LOCATIONS IN THE UK & AT SELECTED LOCATIONS IN SPAIN, AND EVALUATION OF SOLAR THERMAL APPLICATIONS IN THE UK & SPAIN.

Sinatra LI ¹, Pingfan TENG ², Sean CHEN ³ and Julian CHOONG ⁴

¹ MEng, Mechanical; z.li-145@sms.ed.ac.uk

² MEng, Mechanical; p.teng@sms.ed.ac.uk

³ MEng, Mechanical; l.chen-67@sms.ed.ac.uk

⁴ MEng, Mechanical; j.b.c.choong@sms.ed.ac.uk

Abstract: This report compares UK and Spain in the feasibility of solar energy utilization via multiple aspects. In contrast to the UK mainland, Spain has demonstrated unparalleled advantages and leading in multiple aspects. The average national PV potential (GHI) and solar thermal potential (DNI) of Spain are 68.1% and 120.2% respectively higher than the UK's counterparts. The UK also falls behind in terms of the political-economic environment due to Spain's policies offering much higher incentives and tax exemptions while receiving more than €100bil from the EU and its affiliates. This causes a dramatic solar energy capacity difference (35.6GW vs 14.7GW for Spain/UK). This report then uses PVSYST to simulate solar electricity generations of both household and large-scale PV systems. Results show that solar irradiation and shadowing result in the difference in normalized production by PV. Moreover, in terms of solar thermal applications, due to Spain's outstanding DNI, solar thermal is mostly used for concentrating solar power (CSP), whereas in the UK only solar heaters (STH) can be driven. Lastly, the latest state-of-the-arts of STH and CSP are evaluated. Nano-thermal fluids, novel non-concentrating & reflective collectors, and novel thermal energy storages have all demonstrated overall system performance improvements, ranging from 5% to over 35%. Lastly, it is discovered that CSP farms tend to have 25% more specific production than industrial-arrayed PV (IAPV) farms' counterparts.

Keywords: Photovoltaic (PV); Solar Thermal; Solar Energy; Renewable Energy (RES); Feed-in-Tariff (FIT); Smart Export Guarantee (SEG); State of the Art.

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1. Introduction

The rapid society development has been increasing the energy demand, particularly from the source of every sustainable & renewable resource, sun. To participate in so, this project commences with an analysis of solar resources in the UK and Spain. Insiders of PV industry have concluded that the PV capacity of Spain is 242.2% of the UK's counterpart [\[1\]](#) [\[2\]](#) [\[3\]](#), as a consequence of Spain's considerably more attractive feed-in-tariff (FIT) rate, high quota obligation price, generous grants & loans [\[4\]](#) [\[5\]](#). Spain's PV market outlook is extremely positive, as the PV capacity is expected to exceed 82.2GW before 2028, maintaining its first/second largest market in Europe by annual/total installation capacity respectively [\[2\]](#) [\[3\]](#).

Multiple researchers and experimenters have been determining the performance-influencing factors of solar PV panels/systems. Jathar et al [\[5\]](#) discovered that PV electricity generation can easily fluctuate due to the irradiation receiving angle, ambient temperature, panel surface cleanness, and wind speed for cooling. Besides, Fouad et al's research [\[6\]](#)

4.2. Array System

4.2.1. Design Overview

As mentioned in the solar resource sector, Gibraltar has a sufficient solar resource for developing PV solar projects and after identifying the high annual costs of electricity purchase from Ministry of Defense (MoD) [30] (Fig. A3 and Fig. A1), it would be a suitable location to install an Industrial Arrayed Photovoltaic (IAPV) system of about 15MW. Additionally, Gibraltar has its own governing body and regulations that might indicates a positively attitude for developing solar projects. The entire design uses PVSYST to generate the optimal system in terms of the PV module sizing, detailed losses, economic analysis, and the performance result of the system.

4.2.2. Array Type and Sizing

Since the geographical location neglects the impact of the mutual shading effect between rows of arrays after 4 pm and has more than $66808 m^2$ available area for installation. Therefore, in this case, adjusting the pitching distance between rows of arrays would not impact the ground coverage ratio. The optimized orientation for the array to collect the optimal yearly incident irradiation was quickly defined by tuning the tilted and azimuth angles. This result is shown in Fig. 10a Fig. 10b, where the Transposition Factor (TF) is the ratio of yearly incident irradiation on the plane to horizontal irradiation. The corresponding solar path is shown in Fig. 9a.

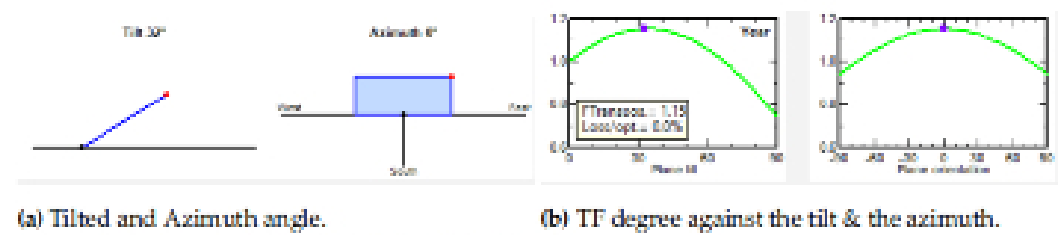


Figure 10. Panel angles and transposition factors (TF).

The decision to design a 15MW capacity solar farm was based on two factors, resolving the monthly units of electricity shortage and promoting green and profitable energy. It was identified through the annual purchase history of Gibraltar's average monthly units of electricity from the Ministry of Defense (MoD) [30]. The mean monthly electricity shortage was computed in Excel from the published purchase history data over a 10-year period[30]. Furthermore, equipping with large-capacity solar power plants also minimizes the dependence on fossil fuel electricity and aligns with the UK's policies [22].

The selected mono-crystalline silicon 580Wp/36v modules by EU manufacturer Sharp have a module efficiency of $22.48\%/m^2$. The selected single phase 3125kW 875-1300 inverters by Sungrow provide 12MW power configuration with the least number of inverters (over 80% of the sized system). All the components are suggested to be arranged as shown in Fig. 11

Fig. 12 clearly shows that the voltage curve of the max. and min. module operating temperature lies between 875 to 1300 V of the inverter operating voltage. Also, both of the voltage curves do not exceed the maximum system voltage of PV or the absolute inverter voltage. Hence the selected modules and inverters have met the PV array sizing conditions.



Figure 11. Diagram of the modules and inverters connection.

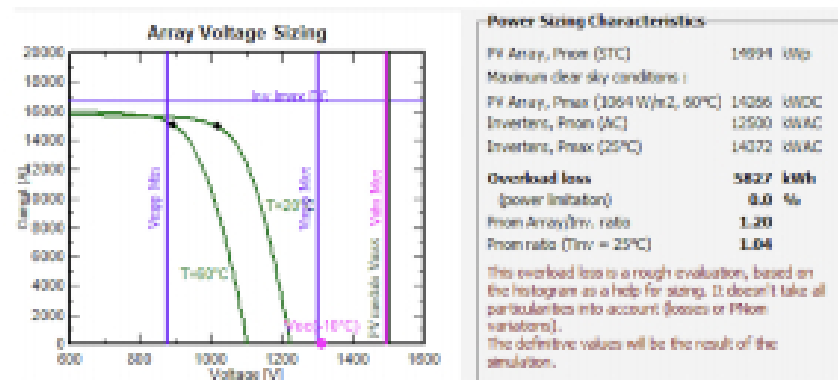


Figure 12. Array sizing conditions.

4.2.3. Losses in Detail

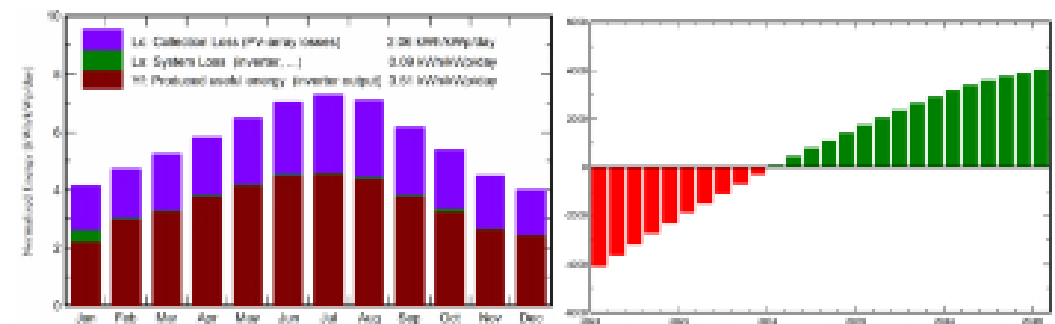
Thermal loss factors were set by the recommended values from PVSYS and from the stimulated results in Fig. 13a which were insignificant, hence no further actions were required to investigate thermal losses. However, most of the losses were due to its site location.

4.2.4. Economic Analysis

The estimated installation, operation cost, UK grants, and average EU bank loan rates were obtained from various sources and then scaled up based on the system size, and FIT was set at £0.12/kWh [44][45][46][47]. The financial result in Fig. 13b shows a potential annual profit of £874,647, LCOE of £0.063/kWh, and would be paid back in 12.3 years.

***Note: The financial report included the aging tool, which was the computation result of the degradation of the PV system over a 25-year period.

4.2.5. Results



(a) Normalized production per installed kWp/day. (b) Cumulative cash flow over 25 years in kGBP. Figure 13. Gibraltar IAPV system production and its estimated long-term revenue.

4.3. Rooftop System

4.3.1. Design Overview

Mainland UK suffers from cloudy weather and generally low solar irradiance compared to all other sites. Norwich was selected due to its relatively sunny weather thanks to its coastal location. The entire design is generated by the PVSYS system. The locations selected are duplex houses with two housing units (referred to as flats) with two stories. These duplex houses are commonly found in Norwich and the UK, allowing the system to be implemented and only requiring an azimuthal angle change. The selected site has a roof slope of 30° which allows for a single, vertical axis tracking PV system. A vertical axis tracking PV system can produce more solar energy compared to a fixed, rooftop PV system.

4.3.2. Array Type and Sizing

The aim of the PV system is not to eliminate the need for electricity but to reduce overall consumption. The sizing heavily depends on the available roof area. Considering the PV arrays will be a vertical axis tracking system, only the roof length is taken into account.

The southern-facing roof has a length of 18 meters, allowing for an 8-meter pitching distance eliminating mutual shading losses. The southern-facing roof is at an azimuthal angle of 28 ° which changes the minimum and maximum azimuthal angle of the PV array. Assuming a 100 ° range (accounting obstruction from the roof when rotating PV arrays), the resulting azimuthal angle is -78 ° to 22 ° as seen in Figure. 14 Due to NOrwich's higher wind conditions with a yearly average wind speed of 16.3kph predominantly from the SW & W direction (49%) [48], a 40 ° tilt panel will require more structural support while only providing a small revenue boost at £1200.

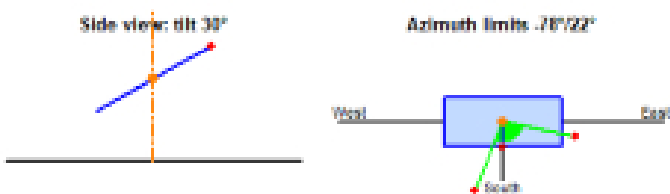


Figure 14. Tilted and Azimuth angle on left and right, without optimisation implemented .

Norwich has a prominent hill northeast of the city center. The location selected is south of the hill so terrain shadows are at a minimum as seen in Figure. 9c. The surrounding area is composed of similarly sized duplex houses with occasional commercial buildings such as grocery shops and gas stations. None of these buildings exceed a height of 10 meters, and therefore will not obstruct any sunlight. Each of the 4-kW systems has its own inverter, this is to reduce the total voltage of the system and maintain inverter health. This can be seen in Figure. 15. The operation conditions of the PV array are within the 218 - 480 V bounds of the inverter.

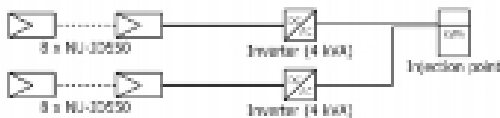


Figure 15. Single line diagram.

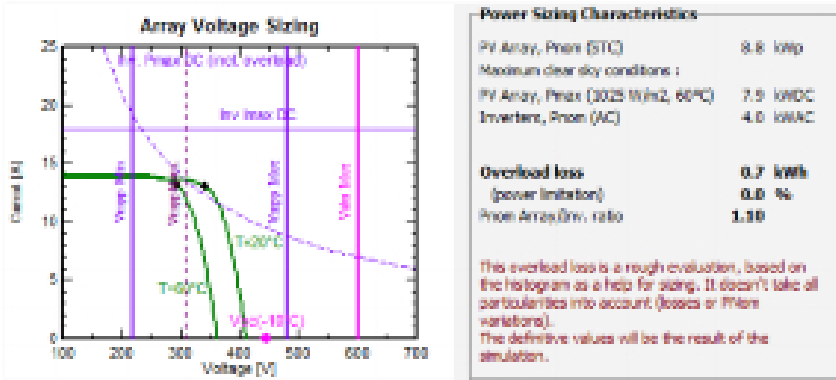


Figure 16. Array sizing.

4.3.3. Detailed losses

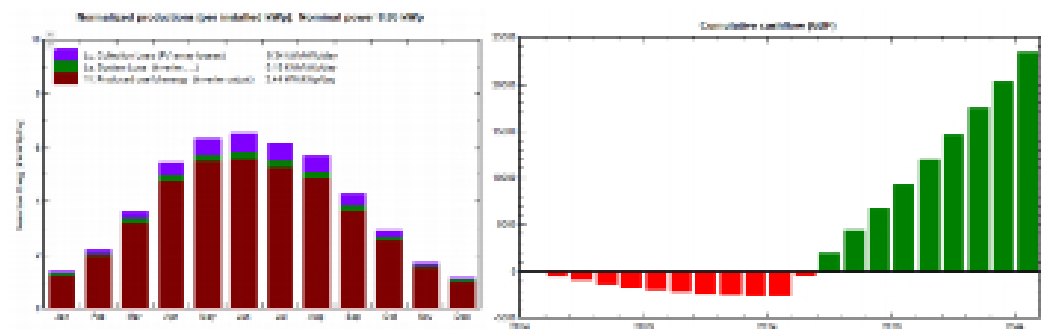
Major loss factors were set by the recommended values from PVSYST and from the stimulated results in Fig. 16 most of the losses were insignificant, hence no further actions were required to investigate this part for this design

4.3.4. Economic analysis

To determine how much money will be made over the project lifespan, we will require the tariff rate and the going consumption tariff rate.

The current SEG rates are determined by electricity supplies. A SEG rate of £0.15/kWh is taken via ScottishPower's website [27], though certain companies such as Octopus offer both fixed and variable tariff [29]. The UK government may raise the tariff rates of the SEG scheme depending on the inflation and current consumption tariff [26]. According to Trading Economics, the average inflation rate for electricity for the past 30 years is 5.58%, and the current consumption tariff rates are £0.24/kWh [30].

The calculation is done based on the worst-case scenario of a constant 5% economic inflation with no adjustments to the SEG rates. Assuming 100 % of the initial installation cost is financed via a loan at 4 %, 10-year fixed interest rate [51]. There will be a self-consumption value double that of the preset template "Household 10 kWh day" within PVSYST due to the house being a duplex type. Assuming air conditioning is only active during summer months (June through August), the average consumption is at 26.3 kWh/day. The system will be paid back in 11.1 years, costing an initial loan of £17,661.12 and resulting in a cumulative profit of £23,418 at a LCOE of £0.1469/kWh.



(a) Normalised production in kWp/day. (b) Cumulative cash flow over 20 years in GBP.

Figure 17. Norwich PV system production and revenue, with the inverter replacement cost distributed.

5. PV Systems in Spain

5.1. Array System

5.1.1. Design Overview

As mentioned in the solar resource sector, Madrid has a great supportive governing body and regulations for developing large-scale PV solar projects. The selected site had taken into consideration of several factors, such as the minimal cloud coverage, neglectable elevation difference, close located grid connection, and least environmental impact. The entire design uses PVSYST to generate the optimal PV system in terms of the PV module sizing, detailed losses, economic analysis, and the performance result of the system.

5.1.2. Array Type and Sizing

The majority of existing PV solar plants in Spain are largely scaled up to 800MW and guaranteed for applicable findings by the government, therefore it is decided to design a grid-connected 150MW PV solar farm close to the Madrid airport where it has an availability of m^2 area for installation.

Unlimited sheds were selected as the array type for this design since the PV arrays are ground mounted, which requires analyzing the ground coverage ratio. The mutual

shading loss shown in Fig. 18 was about 1.8% based on the selected orientation of the sheds (Fig. 18). The obtained shading limit angle was less than 20.0° for the design of a pitching distance of 6.6m and tilted angle of 28° 18. This was verified by the actual solar path, where there was a minimal mutual shading loss during the peak hours(Fig 9b).

Madrid's LAPV will use the same PV modules as Gibraltar has adapted, along with 17x 6874kW 875-1300V inverters from Sungrow (Fig.19) For justifications, refer to Section 4.3.2 In Fig. 20 the array sizing for both selected modules and inverters was met, please refer to Gibraltar's design part for the same justification in the array sizing conditions.

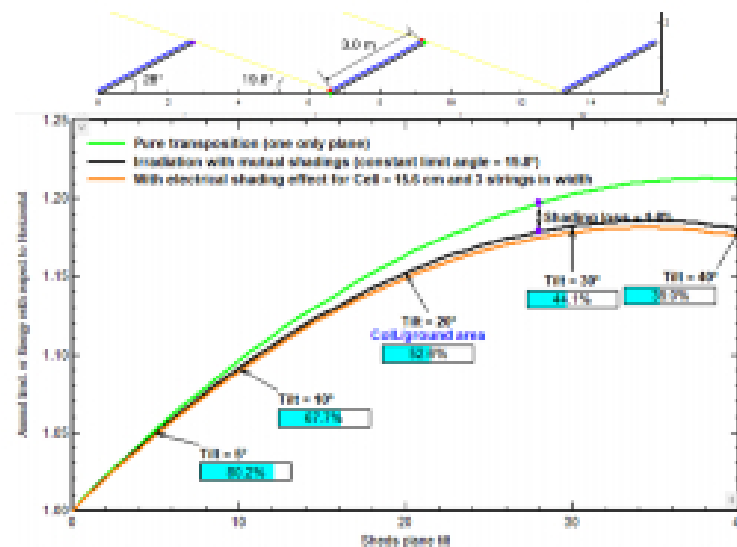


Figure 18. UPPER: Shed angle & setup; LOWER: Shading loss vs Shed setup angle.

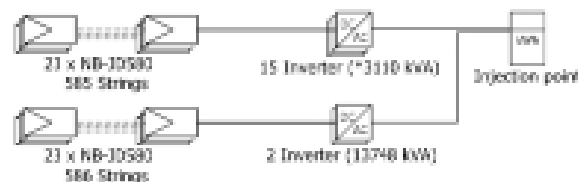


Figure 19. The connection of the selected modules and inverters.

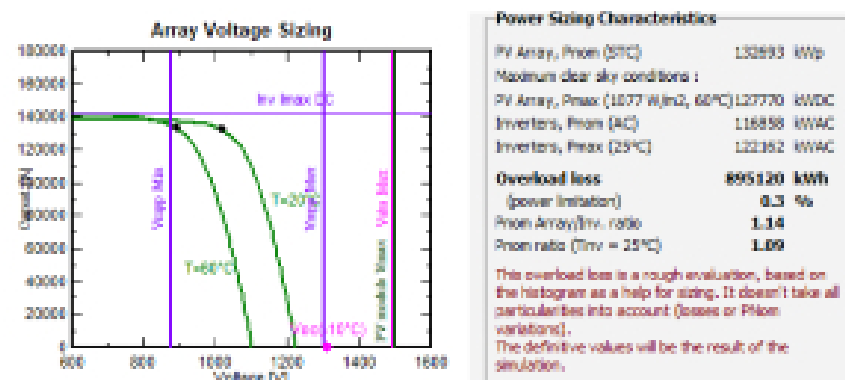


Figure 20. Array sizing conditions.

5.1.3. Losses in Detail

The thermal loss factors of PVSYST were applied in this design and showed an insignificant impact on the overall performance of the PV system (Fig 20), hence no further actions were required to investigate this. The collection losses were caused due to partial mutual shading and average cloud cover in a year.

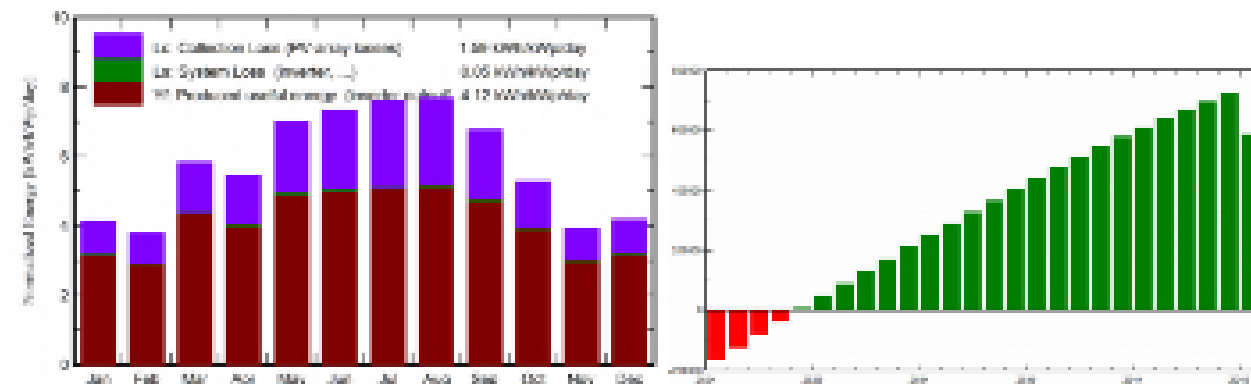
5.1.4. Economic Analysis

Madrid IAPV uses a similar approach as Gibraltar to justify the financial parameters but with a FIT rate of 0.43 EUR/kWh. Besides, per regulations of Green Loan from Re-Power-EU and RIB [7] [52], half of Madrid IAPV's funding can be covered by a 0%-rate loan from the EU commission, and the remaining has to be leased from the national bank, with a loan rate of 4.36% [44] [45] [47] [53]. This is because, for a single project, Green Loan from RE-Power-EU and EIB are only willing to cover up to half of the project costs [53]. On another hand, neither the EU government nor reliable researches reveal the exact interest rate of the Green Loan. A recent study [52] has concluded that the rate is "relatively advantageous [to the loanee]" because EIB only has to "to meet its obligations, to cover its expenses and to constitute a reserve fund", according to the EEC Treaty of Rome (1957).

The purchase price of the land was referred to the average price of 154.4 EUR/ m^2 in Spain [54]. The generated financial result in Fig. 21b shows an estimated average annual profit of 40,434,645 EUR, LCOE of 0.1205 EUR/kWh, and would have been paid back in 11.9 years.

***Note: The financial report included the aging tool, which was the computation result of the degradation of the PV system over a 20-year period.

5.1.5. Results



(a) Normalised production per installed kWp/day. (b) Cumulative cash flow over 25 years in kEUR.
Figure 21. Madrid IAPV system production and its estimated long-term revenue.

Appendix A
Appendix A.1

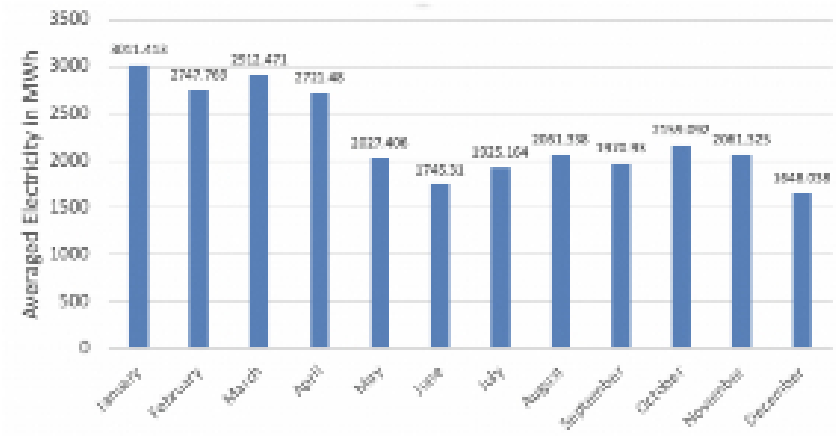


Figure A1. Estimated annual mean units of electricity shortage.

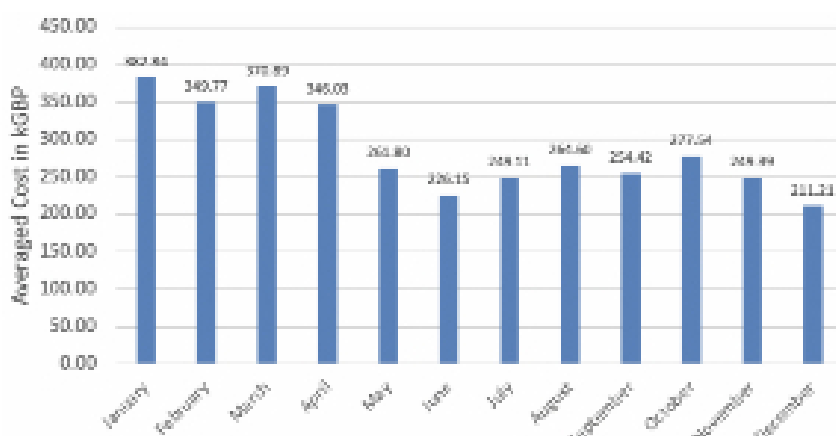


Figure A2. Estimated annual mean electricity imported cost in kGBP.

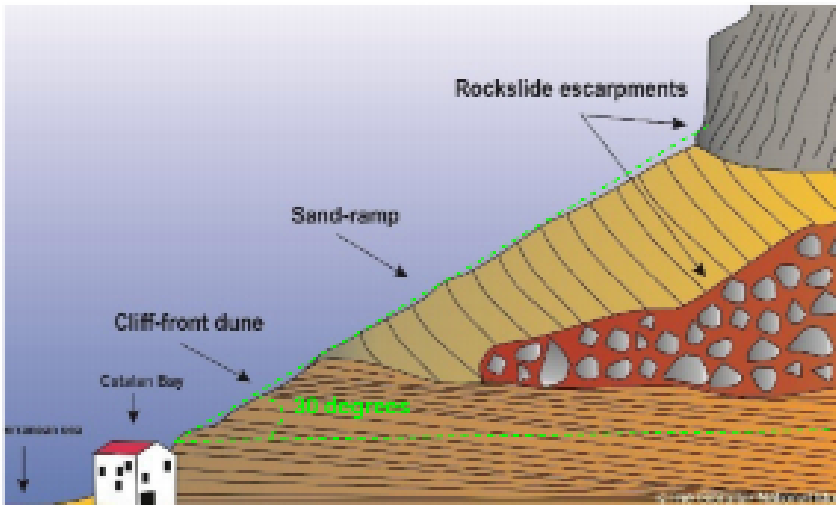


Figure A3. Geographical property of the Sand Dune in Gibraltar, UK. 21